Dara Pooja

cs24btech11021@iith.ac.in | poojasv200620@gmail.com | +91 9573149261 | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.86
Bhashyam Junior college,Gorantla,Guntur	Class XII (Board of Intermediate Education, Andhra Pradesh)	2024	97.60%
Bhashyam High School,Guntur	Class X (Board of Secondary Education, Andhra Pradesh)	2022	94.83%

SKILLS

Languages:

- C++, Python, Java, C, SQL, JavaScript, HTML/CSS, Bash

Developer Tools:

- Linux, Git, CMake, VS Code, Jupyter Notebook, LaTeX

Core Concepts:

- Object-Oriented Programming, Data Structures & Algorithms, Multithreading & Synchronization

PROJECTS

Storage Engine GitHub

(C++, CMake, Git, Linux)

- Built a **persistent disk-backed storage engine** in C++ featuring a **Buffer Pool Manager**, **Disk Storage Manager**, and a **B+ Tree** index for efficient key-value storage.
- Designed the storage layer with **4 KB page-based persistence**, implementing **page allocation**, **free-list based page reuse**, **LRU cache replacement**, **dirty-page flushing**, and **page pinning/unpinning** to optimize memory and disk I/O.
- Developed a **persistent B+ Tree** supporting **insert**, **delete**, **search**, **range scan**, **iterator traversal**, **node split/merge**, and metadata persistence across database restarts.
- Built an **interactive visualizer** with runtime metrics to inspect B+ Tree structure, buffer pool state, cache performance, and disk activity during engine execution.

RISC-V Pipelined Simulator | GitHub

(C++, RISC-V, Computer Architecture)

- Extended an existing **single-cycle RISC-V simulator** into a configurable **5-stage pipelined processor simulator** supporting multiple execution modes with cycle-accurate execution.
- Implemented **hazard detection**, **data forwarding**, **static and dynamic branch prediction**, and a **Branch Target Buffer (BTB)** to resolve data and control hazards.
- Added **floating-point pipeline support**, **instruction scheduling**, **breakpoints**, and **cycle-by-cycle debugging** to improve execution analysis and simulator usability.
- Developed configurable processor modes including **single-cycle**, **basic pipelining**, **hazard detection**, **forwarding**, and **branch prediction** to compare correctness and pipeline performance.

xv6 Kernel Extensions | GitHub

(ANSI C, xv6, RISC-V, Operating Systems)

- Extended the **MIT xv6 (RISC-V)** operating system by implementing custom **system calls**, **process accounting**, and kernel statistics for process management.
- Replaced the default scheduler with a **System-Call-Aware Multi-Level Feedback Queue (SC-MLFQ)** scheduler featuring adaptive priority management, priority boosting, and scheduling statistics.
- Enhanced the virtual memory subsystem by implementing **Clock page replacement**, **page fault handling**, **scheduler-aware page eviction**, and **swap space management**.

Arbitrary Precision Arithmetic Library | GitHub

(Java, Object-Oriented Programming)

- Developed an **arbitrary-precision arithmetic library** in Java using object-oriented design to support integers and floating-point numbers of unbounded size.
- Implemented **addition**, **subtraction**, **multiplication**, and **division** while preserving numerical precision without round-off errors.
- Designed reusable **AInteger** and **AFloat** classes with parsing utilities and a command-line interface for evaluating arithmetic expressions.

RELEVANT COURSES

Computer Science:

Operating Systems I & II*, Database Management Systems I & II*, Computer Architecture*, Algorithms*, Data Structures*,

Compilers I*, Theory of Computation, Software Development Fundamentals*, Introduction to Programming*, Intro to computing

(* includes Lab)

Mathematics:

Probability & Random Variables, Discrete Mathematics, Linear Algebra, Calculus, Statistics

POSITIONS OF RESPONSIBILITY

Event Coordinator, Yoga & Meditation Club (YMC), IIT Hyderabad

- Managed logistics for club events, including venue reservations, faculty coordination, and event communication with students.
- Organized special meditation events and supported smooth event execution.